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(54) **SHOE WITH FOAM PODS AND CHASSIS**

(56) **References Cited**

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U.S. PATENT DOCUMENTS

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6,457,263 B1 10/2002 Rudy
6,665,958 B2 12/2003 Goodwin
6,944,973 B2 9/2005 Goodwin
7,200,955 B2 4/2007 Foxen
7,694,437 B2 4/2010 Hogan
(Continued)

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FOREIGN PATENT DOCUMENTS

CN 102729502 10/2012
CN 105192992 12/2015
(Continued)

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OTHER PUBLICATIONS

International Search Report and Written Opinion for International
Application No. PCT/IB2022/059752, mailed Feb. 2, 2023 (17
pages).

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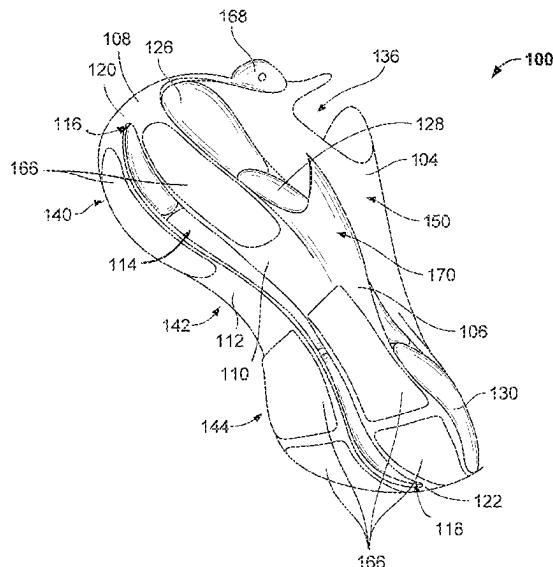
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(57)

ABSTRACT

An article of footwear includes a sole structure coupled with
an upper that defines a forefoot region, a midfoot region, and
a heel region. The sole structure includes a plurality of pods
that each have a pod body formed from a foam material that
optionally has a coating disposed on an outer surface of the
pod body. The sole structure includes a chassis attached to a
bottom surface of each of the plurality of pods. The chassis
includes a plurality of ground contacting regions and an
elongated fork structure. The elongated fork structure
includes a chassis stem, a first chassis leg, and a second
chassis leg that each extend from the chassis stem through
the heel region, the midfoot region, and the forefoot region
of the sole structure. A continuous elongated groove is
positioned between and extends along substantially the
entire length of the first and second chassis leg.

16 Claims, 8 Drawing Sheets



Page 2

(56)	References Cited				2010/0005684	A1 *	1/2010	Nishiwaki	A43B 13/20
U.S. PATENT DOCUMENTS				2011/0185590	A1 *	8/2011	Nishiwaki	A43B 13/187	
7,694,438	B1	4/2010	Christensen	2018/0263336	A1	9/2018	Dekovic	36/28	
8,151,485	B2	4/2012	Hurd	2019/0037964	A1	2/2019	Foxen	36/28	
8,178,022	B2	5/2012	Schindler	2019/0126580	A1 *	5/2019	Paulson	B29C 33/0061	
8,241,450	B2	8/2012	Hensley	2019/0142103	A1	5/2019	Christensen		
8,256,141	B2	9/2012	Christensen	2019/0246738	A1	8/2019	Connell		
8,266,826	B2	9/2012	Cassiday	2019/0261739	A1	8/2019	Taylor		
8,453,344	B2	6/2013	Nishiwaki	2019/0380435	A1	12/2019	Campos		
8,479,412	B2	7/2013	Peyton	2020/0085140	A1 *	3/2020	Hurd	B29D 35/142	
8,490,296	B2	7/2013	Hurd	2020/0196704	A1	6/2020	Dojan		
8,782,924	B2	7/2014	Peyton	2020/0205514	A1	7/2020	VanDomelen		
8,863,407	B2 *	10/2014	Nishiwaki	2020/0260820	A1	8/2020	Youngs		
8,863,408	B2	10/2014	Schindler	2020/0262167	A1	8/2020	Campos, II		
8,919,013	B2	12/2014	Christensen	2020/0315291	A1	10/2020	Tucker		
8,943,709	B2	2/2015	Aveni	2020/0375308	A1	12/2020	Langvin		
8,978,272	B2	3/2015	Hurd	2021/0052034	A1 *	2/2021	Cass	A43B 13/125	
8,978,273	B2	3/2015	Cook	2021/0052037	A1	2/2021	Greenspan		
9,066,556	B2	6/2015	Peyton						
9,066,557	B2	6/2015	Peyton	FOREIGN PATENT DOCUMENTS					
9,089,185	B2 *	7/2015	Nishiwaki	EP	2074901	A1	7/2009		
9,144,266	B2	9/2015	Christensen	EP	1753314	B1	1/2011		
9,179,733	B2	11/2015	Peyton	EP	2197309	B1	11/2012		
9,265,302	B2	2/2016	Peyton	EP	2319343	B1	5/2013		
9,271,544	B2	3/2016	Peyton	EP	2609824	A1	7/2013		
9,289,030	B2	3/2016	Peyton	EP	2369952	B1	12/2013		
9,326,564	B2	5/2016	Peyton	EP	2332432	A4	1/2014		
9,420,848	B2	8/2016	Campos	EP	2568841	B1	7/2014		
9,445,646	B2	9/2016	Cook	EP	2559351	B1	1/2015		
9,456,658	B2	10/2016	Bruce	EP	2227107	B1	4/2015		
9,486,037	B2	11/2016	Cook	EP	2298109	B1	7/2015		
9,516,919	B2	12/2016	Ernst	EP	2197307	B1	8/2015		
9,521,877	B2	12/2016	Campos, II	EP	2227106	B1	2/2016		
9,629,414	B2	4/2017	Elder	EP	2506735	B1	7/2016		
9,750,307	B2	9/2017	Campos, II	EP	2897483	B1	12/2016		
9,801,428	B2	10/2017	Taylor	EP	2937008	B1	7/2017		
9,894,959	B2	2/2018	Taylor	EP	2793630	B1	8/2017		
9,907,357	B2	3/2018	Youngs	EP	2897484	B1	8/2017		
9,913,511	B2	3/2018	Peyton	EP	2227108	B1	11/2017		
9,981,437	B2	5/2018	Campos, II	EP	2910140	B1	12/2017		
9,987,814	B2	6/2018	Campos, II	EP	2979566	B1	12/2017		
10,010,136	B2	7/2018	Shaffer	EP	2229067	B1	2/2018		
10,098,410	B2	10/2018	Cook	EP	2929791	B1	3/2018		
10,111,492	B2	10/2018	Foxen	EP	3338582	A1	6/2018		
10,178,891	B2	1/2019	Christensen	EP	2897482	B1	10/2018		
10,244,821	B2	4/2019	Elder	EP	2764787	B1	11/2018		
10,271,613	B2	4/2019	Campos	EP	3243400	B1	3/2019		
10,299,536	B2	5/2019	Ernst	EP	3185712	B1	6/2019		
10,327,507	B2	6/2019	Taylor	EP	3692852	A1	8/2020		
10,441,029	B2	10/2019	Campos, II	EP	3692854	A1	8/2020		
10,477,919	B2	11/2019	Taylor	EP	3721739	A1	10/2020		
10,624,417	B2	4/2020	Dojan	EP	3429391	B1	11/2020		
10,675,828	B2	6/2020	Campos, II	EP	3095344	B1	12/2020		
10,702,016	B2	7/2020	Youngs	EP	3250073	B1	12/2020		
10,729,206	B2	8/2020	Campos, II	EP	3597067	B1	2/2021		
10,743,609	B2	8/2020	Meschter	EP	3785562	A1	3/2021		
10,758,011	B2 *	9/2020	Bock	WO	2015175605	A1	11/2015		
10,849,387	B2	12/2020	Bruce	WO	2017053692	A1	3/2017		
10,856,612	B2	12/2020	Heard	WO	2017160730	A1	9/2017		
10,897,958	B2	1/2021	Peyton	WO	2019157244	A1	8/2019		
10,939,726	B2	3/2021	Scofield	WO	2020099439	A1	5/2020		
11,033,079	B2 *	6/2021	Bock	WO	2020142429	A1	7/2020		
2004/0049946	A1 *	3/2004	Lucas	WO	2020206240	A1	10/2020		
2006/0117599	A1 *	6/2006	Deem	WO	2020243385	A1	12/2020		
2006/0248750	A1	11/2006	Rosenberg	WO	2021041269	A1	3/2021		
2009/0013558	A1 *	1/2009	Hazenberg	WO	WO-2023067439	A1 *	4/2023	A43B 13/122	

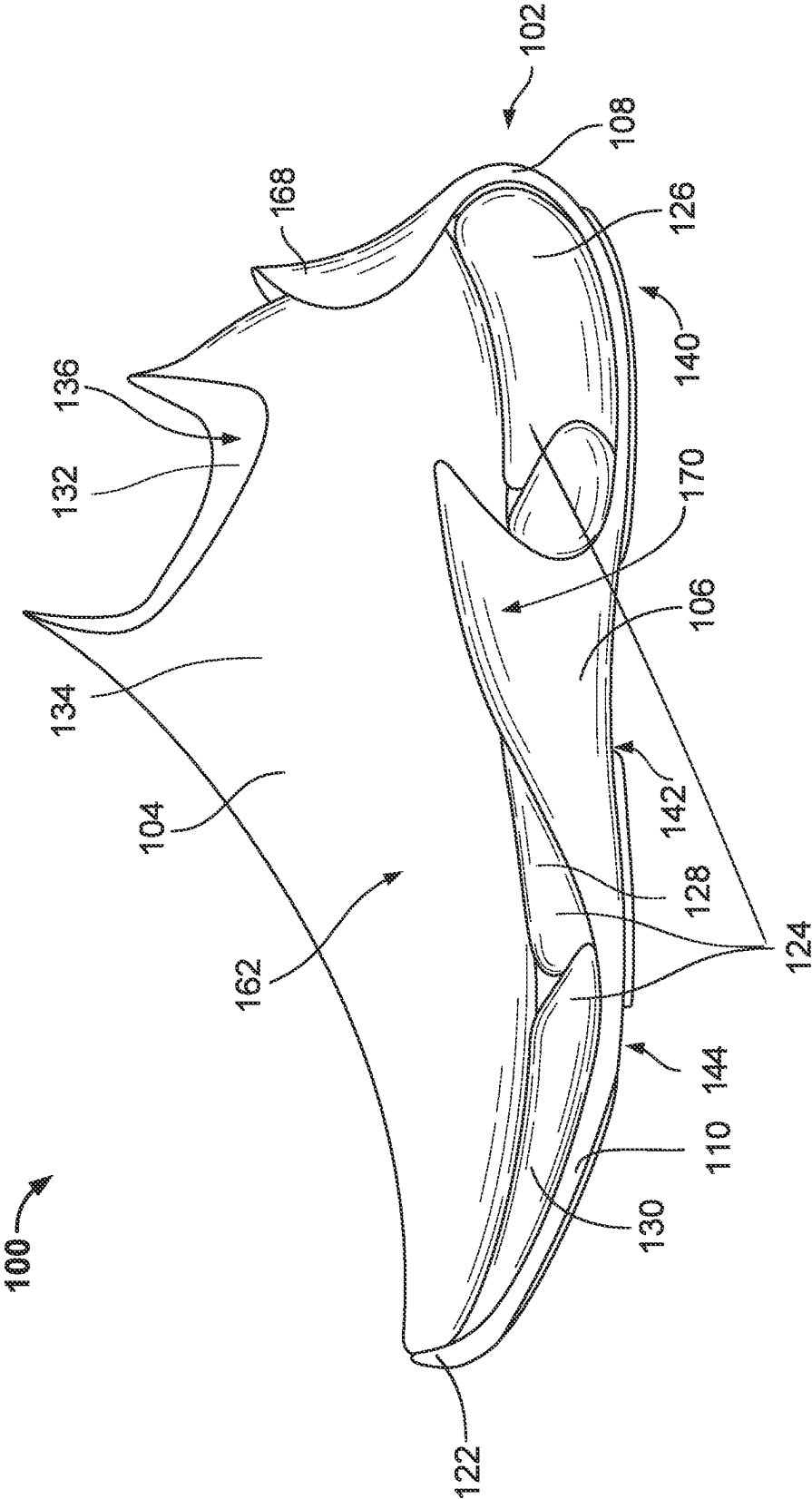


FIG. 1

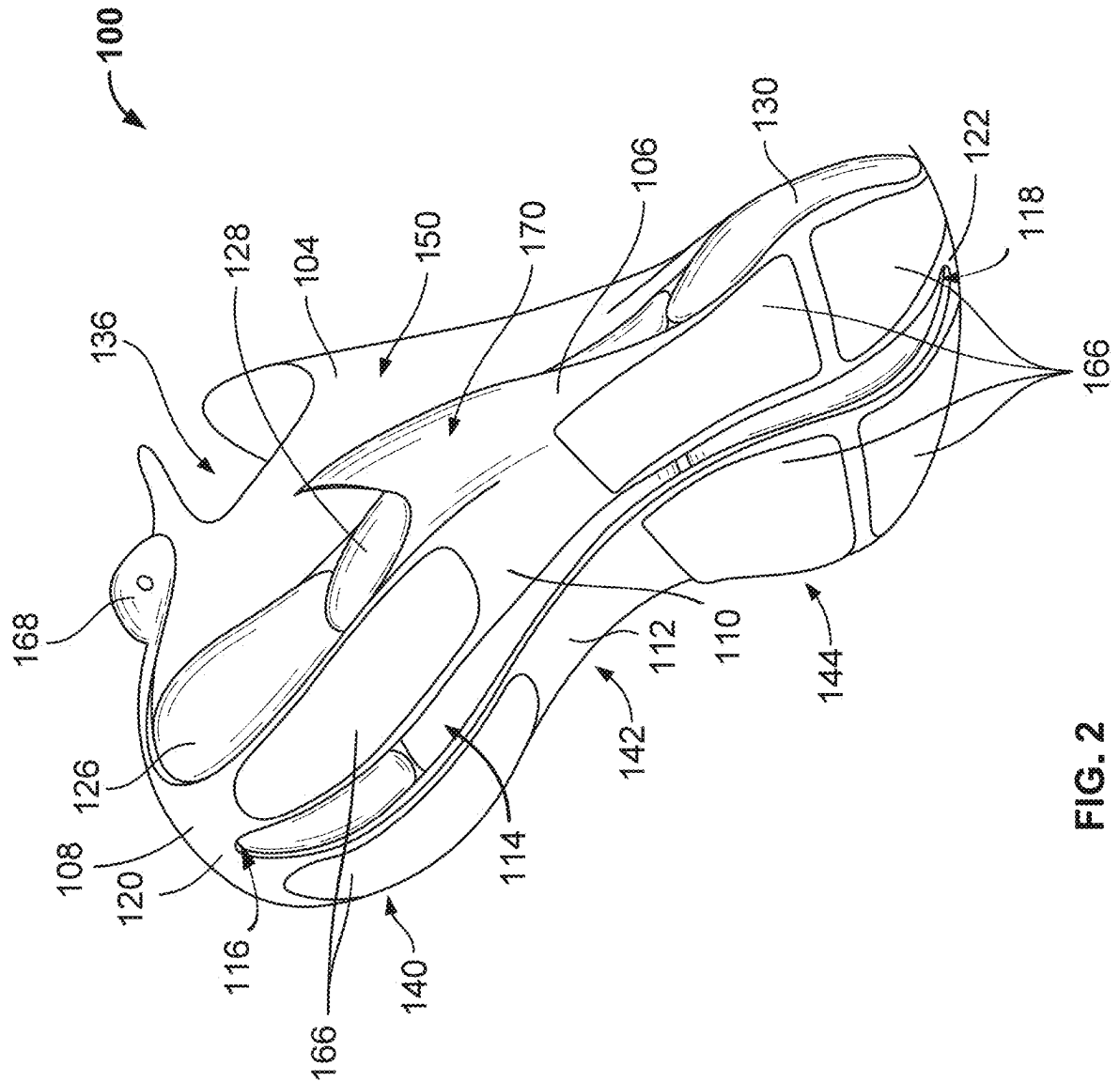
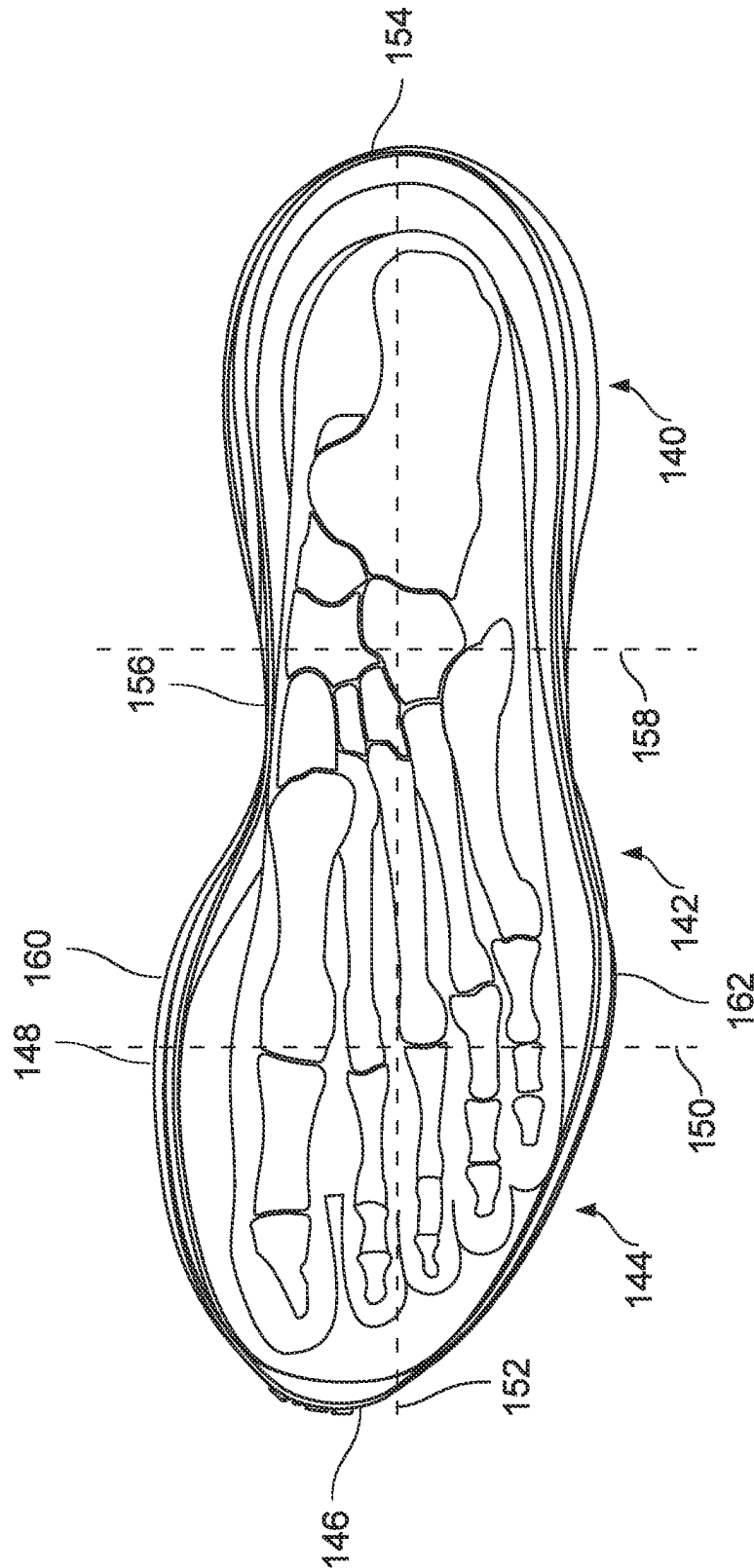


FIG. 2



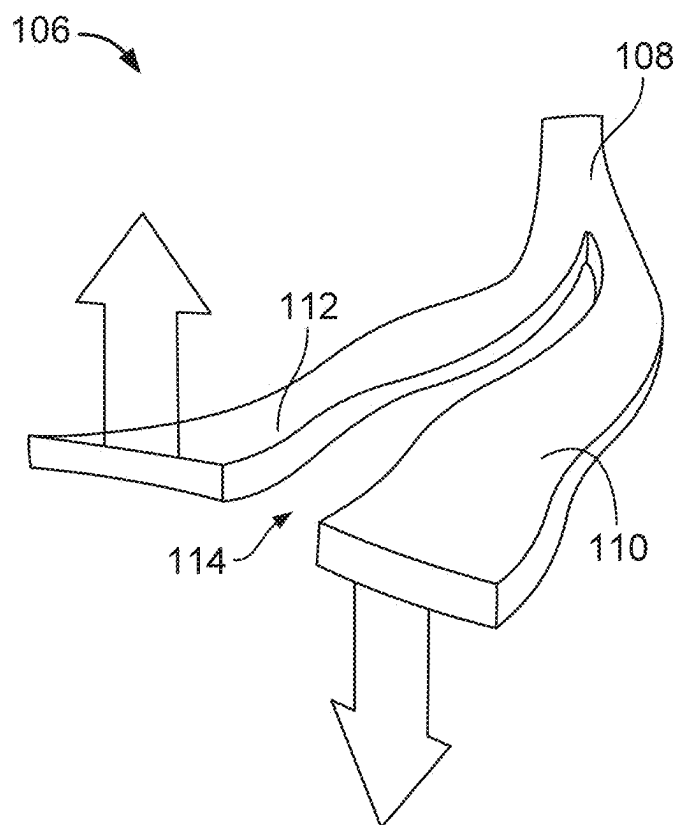


FIG. 4

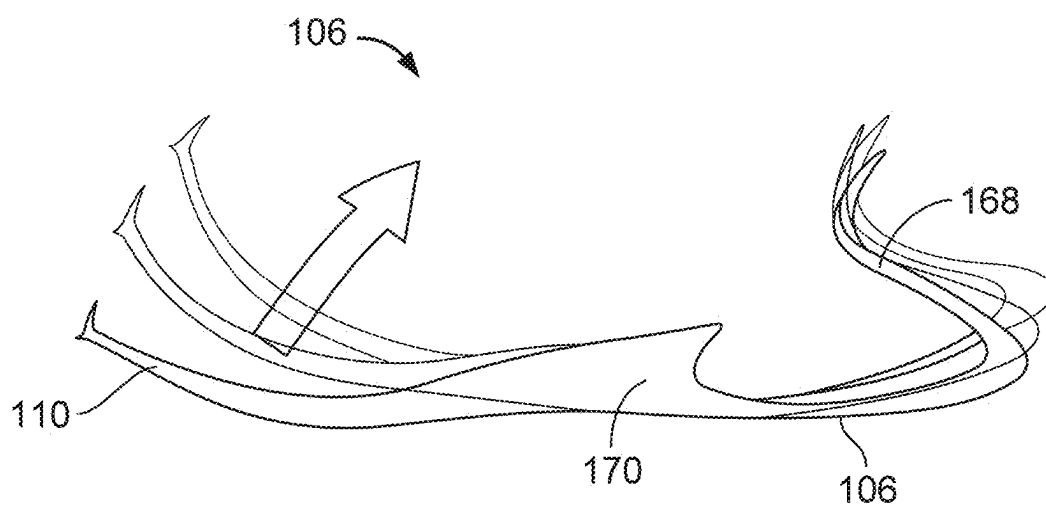


FIG. 5

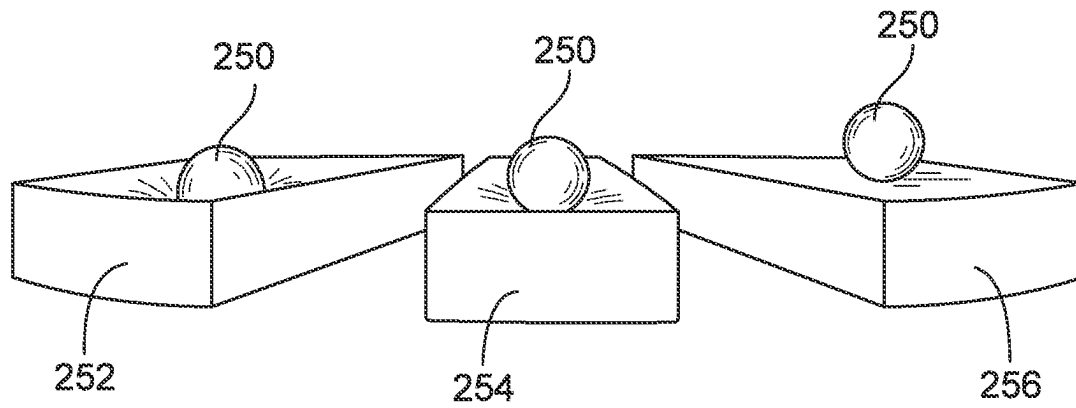


FIG. 6

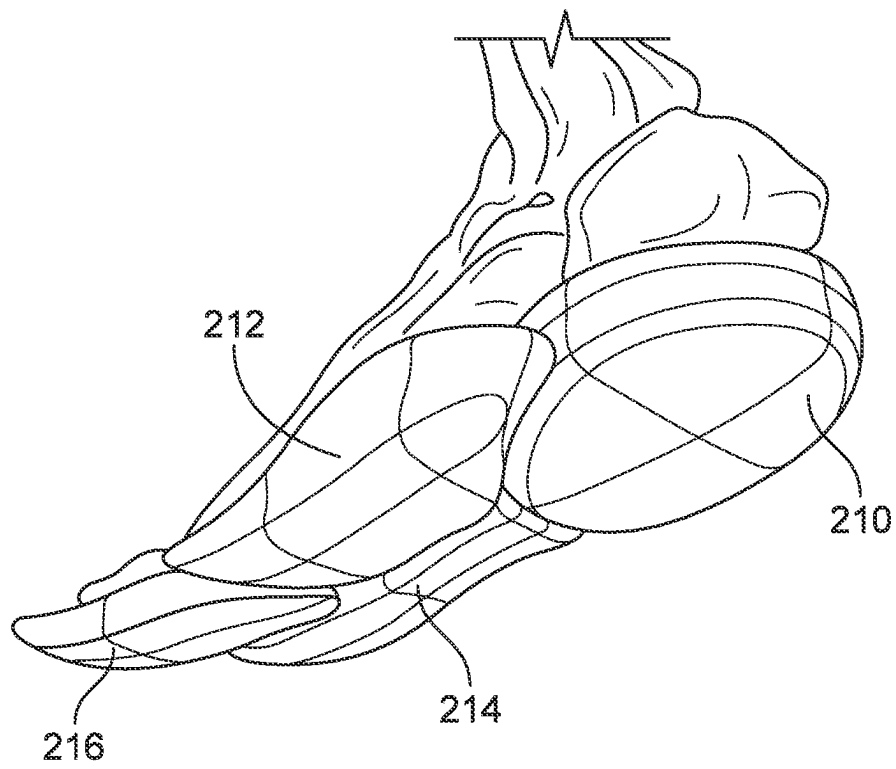
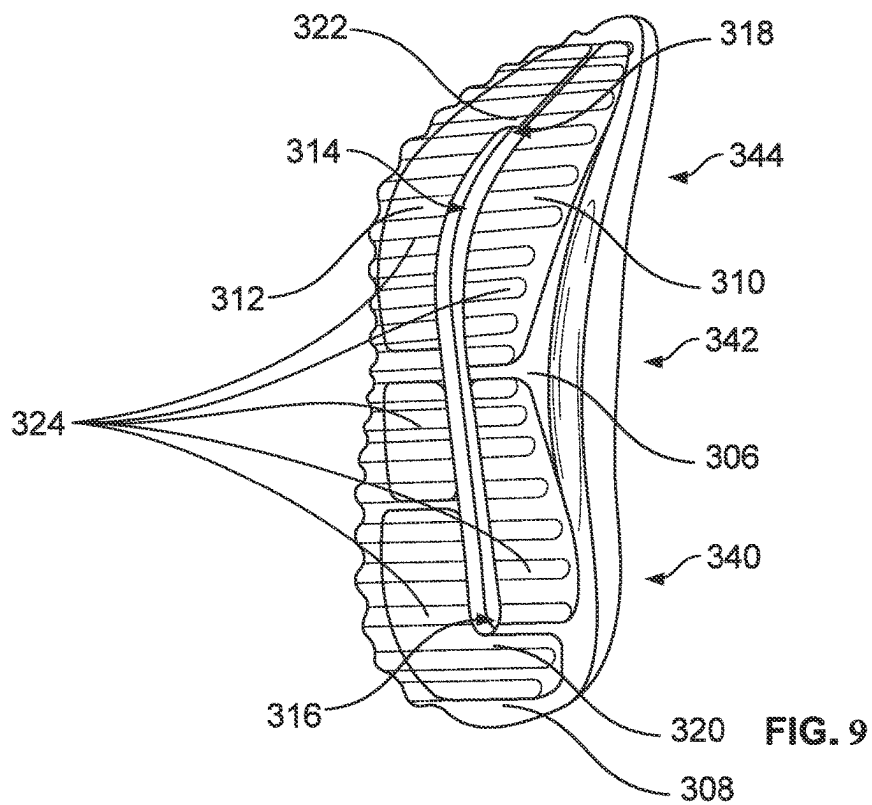
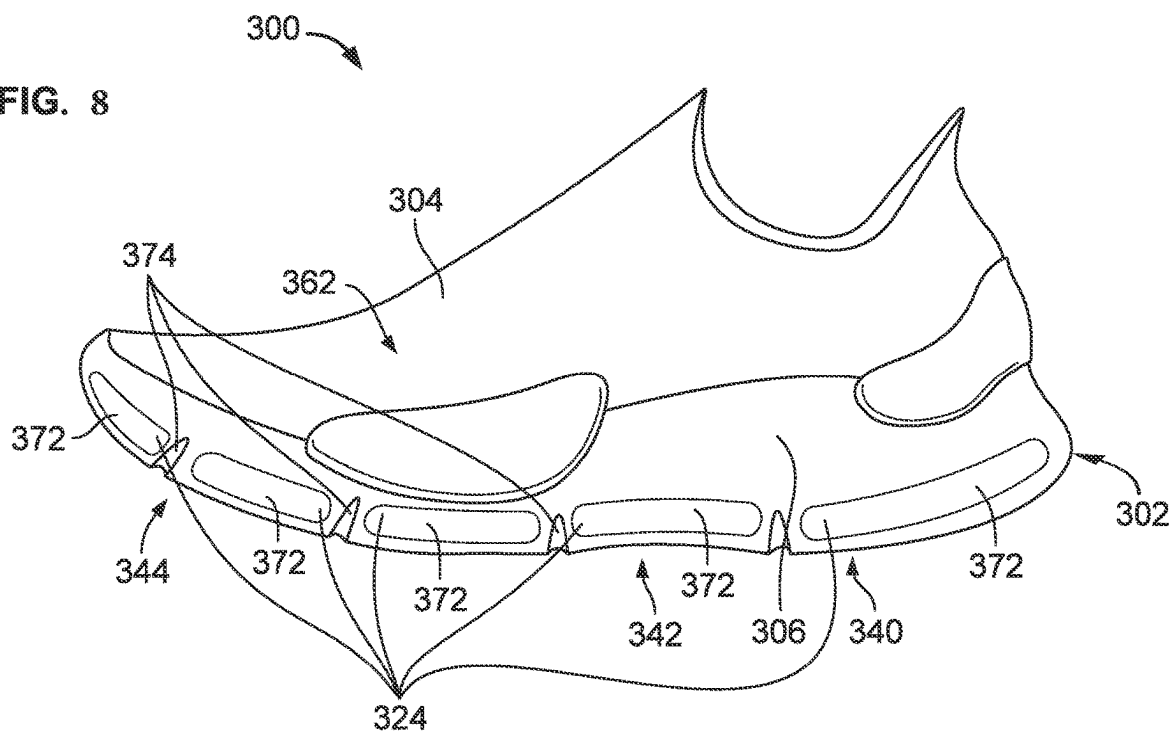


FIG. 7

FIG. 8



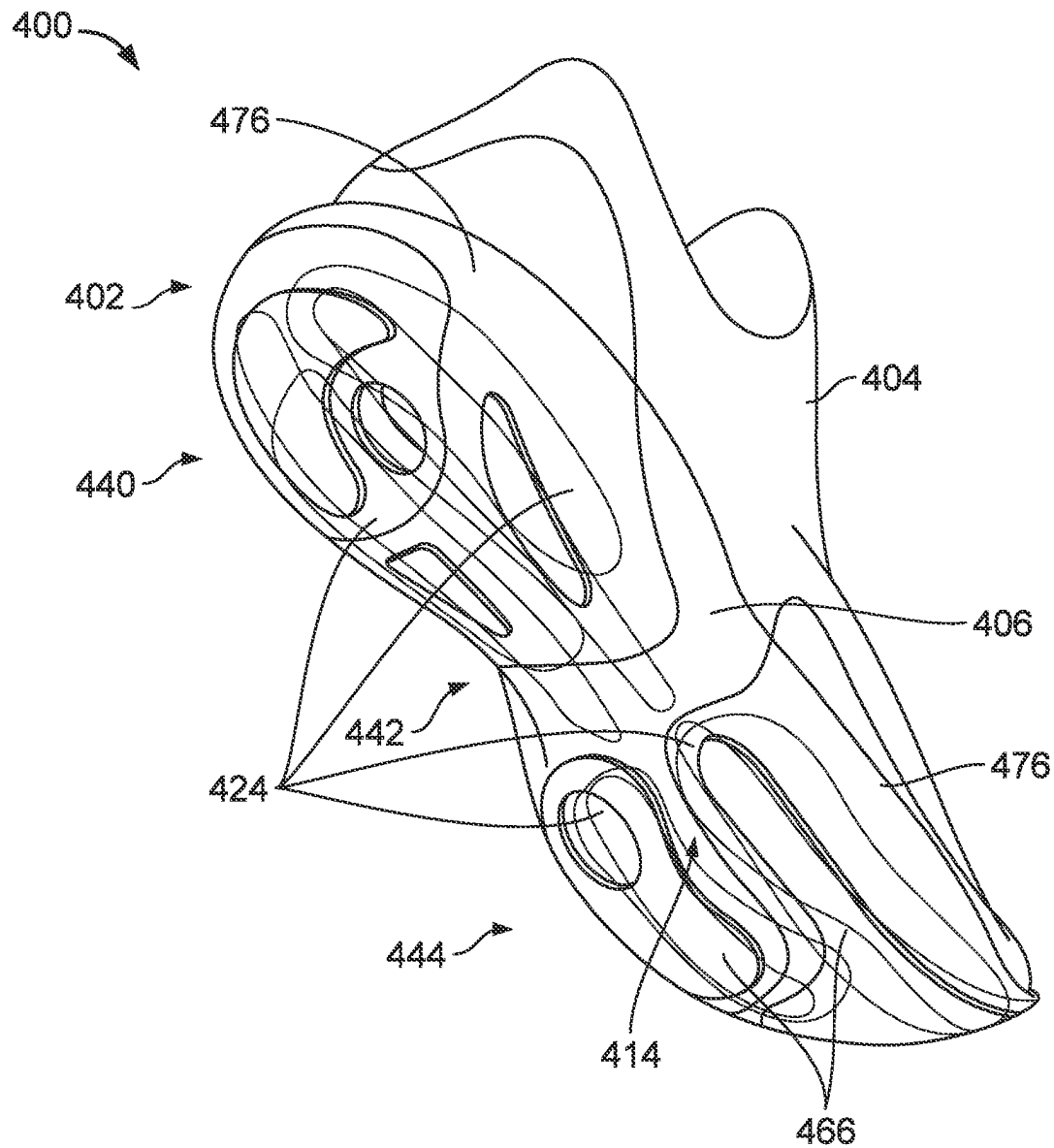


FIG. 10

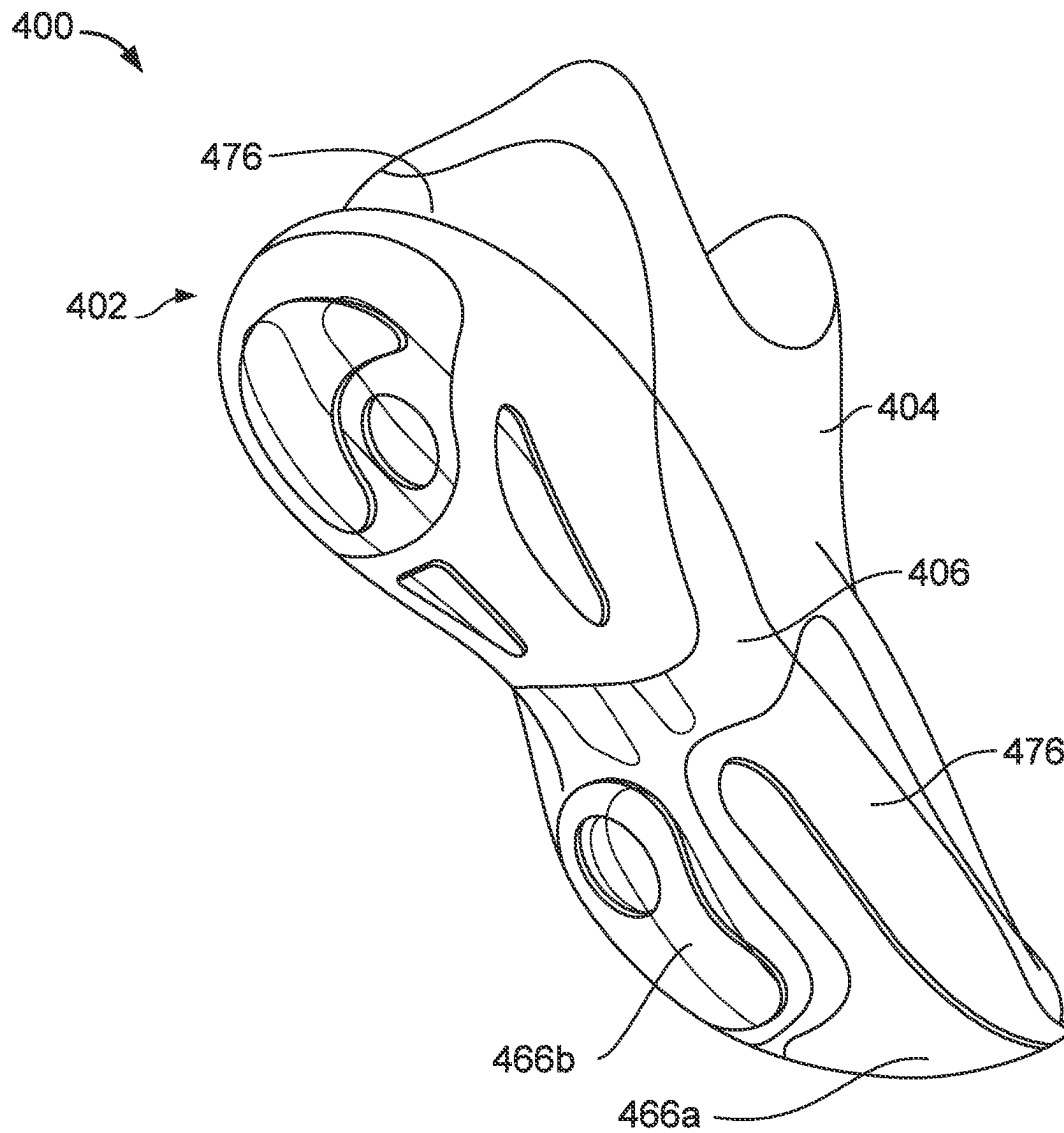


FIG. 11

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SHOE WITH FOAM PODS AND CHASSIS**CROSS REFERENCE TO RELATED APPLICATIONS**

This application claims the benefit of and priority to U.S. Provisional App. No. 63/256,874, filed on Oct. 18, 2021, entitled "SHOE WITH FOAM PODS AND CHASSIS", which is herein incorporated by reference in its entirety.

BACKGROUND**1. Field of the Disclosure**

The present disclosure relates generally to an article of footwear.

2. Description of the Background

Many conventional shoes or other articles of footwear generally comprise an upper and a sole attached to a lower end of the upper. Conventional shoes further include an internal space, i.e., a void or cavity, which is created by interior surfaces of the upper and sole, that receives a foot of a user before securing the shoe to the foot. The sole is attached to a lower surface or boundary of the upper and is positioned between the upper and the ground. As a result, the sole typically provides stability and cushioning to the user when the shoe is being worn. In some instances, the sole may include multiple components, such as an outsole, a midsole, and an insole. The outsole may provide traction to a bottom surface of the sole, and the midsole may be attached to an inner surface of the outsole, and may provide cushioning or added stability to the sole. For example, a sole may include a particular foam material that may increase stability at one or more desired locations along the sole, or a foam material that may reduce stress or impact energy on the foot or leg when a user is running, walking, or engaged in another activity. The sole may also include additional components, such as plates, embedded with the sole to increase the overall stiffness of the sole and reduce energy loss during use.

The upper generally extends upward from the sole and defines an interior cavity that completely or partially encases a foot. In most cases, the upper extends over the instep and toe regions of the foot, and across medial and lateral sides thereof. Many articles of footwear may also include a tongue that extends across the instep region to bridge a gap between edges of medial and lateral sides of the upper, which define an opening into the cavity. The tongue may also be disposed below a lacing system and between medial and lateral sides of the upper, to allow for adjustment of shoe tightness. The tongue may further be manipulable by a user to permit entry or exit of a foot from the internal space or cavity. In addition, the lacing system may allow a user to adjust certain dimensions of the upper or the sole, thereby allowing the upper to accommodate a wide variety of foot types having varying sizes and shapes.

The upper may comprise a wide variety of materials, which may be chosen based on one or more intended uses of the shoe. The upper may also include portions comprising varying materials specific to a particular area of the upper. For example, added stability may be desirable at a front of the upper or adjacent a heel region so as to provide a higher degree of resistance or rigidity. In contrast, other portions of

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a shoe may include a soft woven textile to provide an area with stretch-resistance, flexibility, air-permeability, or moisture-wicking properties.

However, while many currently-available shoes have varying features related to the above-noted properties, many shoes, and the sole structures thereof, may be further optimized to provide targeted support to a user's foot to aid in stability while running, walking, or engaging in strenuous athletic activities. Additionally, many shoes, and their sole structures, may be further optimized to provide targeted support to a user's foot to reduce energy dissipation and thereby increase the efficiency of a user during physical activity, such as running. Moreover, many shoes, and their sole structures, may be further optimized to provide additional support and cushioning to a user's bones and tissue in the user's foot. Still further, many shoes, and their sole structures, may be optimized to provide protection for a user's tissue, bones, and ligaments from excessive loads.

Therefore, articles of footwear having features providing such effects across areas of the foot are desired. These and other deficiencies with the prior art are outlined in the following disclosure.

SUMMARY

An article of footwear, as described herein, may have various configurations. The article of footwear may have an upper and a sole structure that is coupled with the upper. The sole structure may define a forefoot region, a midfoot region, and a heel region. The sole structure may include a plurality of pods. Each pod may have a pod body and may also optionally have a coating that is disposed on an outer surface of the pod body. The sole structure may also include a chassis that is attached to a bottom surface of each pod in the plurality of pods. The pod body of each pod in the plurality of pods may be formed from a foam material. The chassis may include a plurality of ground contacting regions. The chassis may have an elongated fork structure. The elongated fork includes a chassis stem, a first chassis leg that extends from the chassis stem through the heel region, the midfoot region, and the forefoot region of the sole structure, and a second chassis leg that also extends from the chassis stem through the heel region, the midfoot region, and the forefoot region. The elongated fork structure also has a continuous elongated groove that is positioned between the first chassis leg and the second chassis leg. The continuous elongated groove extends along substantially the entire length of the first chassis leg and the second chassis leg.

The article of footwear, as described herein, may have an upper and a sole structure that is coupled with the upper. The sole structure may define a forefoot region, a midfoot region, and a heel region. The sole structure may include a plurality of pods. Each pod may have a pod body and may also optionally have a coating that is disposed on an outer surface of the pod body. The sole structure may also include a chassis that is attached to a bottom surface of each pod in the plurality of pods. The pod body of each pod in the plurality of pods is formed from a foam material. The foam material in the pod body of at least one pod in the plurality of pods may have a density that is different than the density of the foam material in the pod body of at least one other pod in the plurality of pods. The chassis may include a plurality of ground contacting regions. The chassis may not be attached to the upper.

In another embodiment, the article of footwear may have an upper and a sole structure that is coupled with the upper. The sole structure may define a forefoot region, a midfoot

region, and a heel region. The sole structure may include a plurality of pods. Each pod may have a pod body and may also optionally have a coating that is disposed on an outer surface of the pod body. The sole structure may also include a chassis that is attached to a bottom surface of each pod in the plurality of pods. The pod body of each pod in the plurality of pods is formed from a foam material. The foam material in the pod body of at least one pod in the plurality of pods may have a density that is different than the density of the foam material in the pod body of at least one other pod in the plurality of pods. The chassis may encapsulate each pod in the plurality of pods.

Other aspects of the articles of footwear described herein, including features and advantages thereof, will become apparent to one of ordinary skill in the art upon examination of the figures and detailed description herein. Therefore, all such aspects of the articles of footwear are intended to be included in the detailed description and this summary.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 a lateral side view of an article of footwear configured as a left shoe that includes an upper and a sole structure, as discussed herein;

FIG. 2 is a bottom perspective view of the shoe of FIG. 1;

FIG. 3 is a top plan view of the shoe of FIG. 1, with the upper removed and a user's skeletal foot structure overlaid thereon;

FIG. 4 is a front view of a portion of a chassis of the shoe of FIG. 1;

FIG. 5 is a lateral side motion view of a portion of a chassis of the shoe of FIG. 1;

FIG. 6 is a representation of three types of foam material that may be suitable for use in an article of footwear as discussed herein;

FIG. 7 is a bottom perspective view of pods according to an embodiment of the present disclosure, shown in arrangement contacting a user's skeletal foot structure;

FIG. 8 is a lateral side view of the shoe of another article of footwear configured as a left shoe that includes an upper and a sole structure, as discussed herein;

FIG. 9 is a bottom perspective view of a portion of a chassis, according to an embodiment of the present disclosure;

FIG. 10 is a bottom perspective view of an article of footwear configured according to an embodiment of the present disclosure; and

FIG. 11 is a bottom perspective view of yet an article of footwear configured according to an embodiment of the present disclosure.

DETAILED DESCRIPTION OF THE DRAWINGS

The following discussion and accompanying figures disclose various embodiments or configurations of a shoe having an upper and a sole structure. Although embodiments are disclosed with reference to a sports shoe, such as a running shoe, tennis shoe, basketball shoe, etc., concepts associated with embodiments of the shoe may be applied to a wide range of footwear and footwear styles, including cross-training shoes, football shoes, golf shoes, hiking shoes, hiking boots, ski and snowboard boots, soccer shoes and cleats, walking shoes, and track cleats, for example. Concepts of the shoe may also be applied to articles of footwear that are considered non-athletic, including dress shoes, sandals, loafers, slippers, and heels.

The term "about," as used herein, refers to variations in the numerical quantity that may occur, for example, through typical measuring and manufacturing procedures used for articles of footwear or other articles of manufacture that may include embodiments of the disclosure herein; through inadvertent error in these procedures; through differences in the manufacture, source, or purity of the ingredients used to make the compositions or mixtures or carry out the methods; and the like. Throughout the disclosure, the terms "about" and "approximately" refer to a range of values $\pm 5\%$ of the numeric value that the term precedes.

The present disclosure is directed to an article of footwear or specific components of the article of footwear, such as an upper or a sole or a sole structure. The sole structure defines a forefoot region, a midfoot region, and a heel region. The sole structure is coupled with the upper. The sole structure includes a plurality of pods and a chassis. The plurality of pods are connected to the upper. More specifically, the top surface of each pod is connected to the bottom surface of the upper. The chassis is connected to the plurality of pods. More specifically, the bottom surface of each pod is connected to at least the top surface of the chassis. In an embodiment, the chassis is not connected directly to the upper.

While only a single shoe is depicted, i.e., a shoe that is worn on a left foot of a user, it should be appreciated that the concepts disclosed herein are applicable to a pair of shoes (not shown), which includes a left shoe and a right shoe that may be sized and shaped to receive a left foot and a right foot of a user, respectively. For ease of disclosure, however, a single shoe will be referenced to describe aspects of the disclosure, but the disclosure below with reference to the article of footwear is applicable to both a left shoe and a right shoe. However, in some embodiments there may be differences between a left shoe and a right shoe other than the left/right configuration. Further, a left shoe may include one or more additional elements that a right shoe does not include, or vice versa.

Referring initially to FIGS. 1 and 2, the article of footwear 100 comprises sole structure 102 and an upper 104. The upper 104 will be discussed in more detail below. The sole structure 102 is coupled with the upper 104. The sole structure 102 includes a chassis 106 having a chassis stem 108, a first chassis leg 110, a second chassis leg 112 (shown in FIG. 2), and a continuous elongated groove 114. The continuous elongated groove 114 has a first end 116, a second end 118, a heel joint 120, and a toe joint 122. The sole structure 102 also includes a plurality of pods 124. The plurality of pods 124 includes a first pod 126, a second pod 128, and third pod 130. The chassis 106 is coupled with the plurality of pods 124. The plurality of pods 124 is connected directly with the upper 104. The chassis 106 may be connected directly to the upper 104. However, in some embodiments, it is preferred that the chassis 106 is not connected directly to the upper 104, but rather the chassis 106 is only indirectly coupled with the upper 104 by way of attachment between the chassis 106 and the plurality of pods 124 and further attachment between the plurality of pods 124 and the upper 104. When fully assembled, the upper 104 includes an interior surface 132, an exterior surface 134, and an interior cavity 136. The interior surface 132 faces inward and generally defines the interior cavity 136, and the exterior surface 134 of the upper 104 faces outward and generally defines an outer perimeter or boundary of the upper 104. As will be discussed in greater detail below, the pods in the plurality of pods 124 may be customized so that a given

plurality of pods **124** may have specific properties that are tailored to meet the needs of a given user.

Referring now to FIG. 3, the article of footwear **100** has a heel region **140**, a midfoot region **142**, and a forefoot region **144**. The forefoot region **144** generally corresponds with portions of the article of footwear **100** that encase portions of the foot that includes the toes, the ball of the foot, and joints connecting the metatarsals with the toes or phalanges. The midfoot region **142** is proximate and adjoining the forefoot region **144**, and generally corresponds with portions of the article of footwear **100** that encase the arch of the foot, along with the bridge of a foot. The heel region **140** is proximate and adjoining the midfoot region **142** and generally corresponds with portions of the article of footwear **100** that encase rear portions of the foot, including the heel or calcaneus bone, the ankle, or the Achilles tendon.

The forefoot region **144** extends from a toe end **146** to a widest portion **148** of the article of footwear **100**. The widest portion **148** is defined or measured along a first line **150** that is perpendicular with respect to a longitudinal axis **152** that extends from a distal portion of the toe end to a distal portion of a heel end **154**, which is opposite the toe end. The midfoot region **142** extends from the widest portion **148** to a thinnest portion **156** of the article of footwear **100**. The thinnest portion **156** of the article of footwear **100** is defined as the portion of the article of footwear **100** having the least width, as measured along a second line **158** that is perpendicular with respect to the longitudinal axis **152**. The heel region **140** extends from the thinnest portion **156** to the heel end **154** of the article of footwear **100**.

The heel region **140**, the midfoot region **142**, the forefoot region **144**, a medial side **160**, and a lateral side **162** define boundaries or areas of the article of footwear **100**, and generally characterize sections of the article of footwear **100**. Certain aspects of the disclosure may refer to portions or elements that are coextensive with one or more of the heel region **140**, the midfoot region **142**, the forefoot region **144**, the medial side **160**, or the lateral side **162**. Further, both the upper **104** and the sole structure **102** may be characterized as having portions within the heel region **140**, the midfoot region **142**, the forefoot region **144**, or along the medial side **160** or the lateral side **162**. Therefore, the upper **104** and the sole structure **102**, may include portions thereof that are disposed within the heel region **140**, the midfoot region **142**, the forefoot region **144**, or along the medial side **160** or the lateral side **162**.

When a user is wearing the article of footwear **100** (configured as a pair of shoes), the lateral side **162** corresponds with an outside-facing portion of the article of footwear **100** while the medial side **160** corresponds with an inside-facing portion of the article of footwear **100**. As such, a left shoe and a right shoe have opposing lateral sides **162** and medial sides **160**, such that the medial sides **160** are closest to one another when a user is wearing the shoes, while the lateral sides **162** are the sides that are farthest from one another. As will be discussed in greater detail below, the medial side **160** and the lateral side **162** adjoin one another along a longitudinal central plane or axis **152** at opposing, distal ends of the article of footwear **100**. As will also be further discussed herein, the longitudinal central plane or axis **152** may demarcate a central, intermediate axis between the medial side **160** and the lateral side **162** of the article of footwear **100**. Put differently, the longitudinal plane or axis **152** may extend between a toe end **146** of the article of footwear **100** and a heel end **154** of the article of footwear **100**. The longitudinal axis **152** may continuously define a

middle of the sole structure **102**, or the upper **104** of the article of footwear **100**, i.e., the longitudinal plane or axis **152** may be a straight axis extending from the heel end **154** of the heel region **140** to the toe end **146** of the forefoot region **144**.

Still referring to FIG. 3, the medial side **160** begins at the toe end **146** and bows outward along the forefoot region **144** toward the midfoot region **142**. At the first line **150**, the medial side **160** bows inward, toward the central, longitudinal axis **152**. The medial side **160** extends from the first line **150**, i.e., the widest portion **148**, toward the second line **158**, i.e., the thinnest portion **156**, entering into the midfoot region **142** upon crossing the first line **150**. After reaching the second line **158**, the medial side **160** bows outward, away from the longitudinal, central axis **152**, at which point the medial side **160** extends into the heel region **140**, i.e., upon crossing the second line **158**. The medial side **160** then bows outward and then inward toward the heel end **154**, and terminates at a point where the medial side **160** meets the longitudinal, center axis **152**.

The lateral side **162** also begins at the toe end **146** and bows outward along the forefoot region **144** toward the midfoot region **142**. The lateral side **162** reaches the first line **150**, at which point the lateral side **162** bows inward, toward the longitudinal, central axis **152**. The lateral side **162** extends from the first line **150**, i.e., the widest portion **148**, toward the second line **158**, i.e., the thinnest portion **156**, entering into the midfoot region **142** upon crossing the first line **150**. After reaching the second line **158**, the lateral side **162** bows outward, away from the longitudinal, central axis **152**, at which point the lateral side **162** extends into the heel region **140**, i.e., upon crossing the second line **158**. The lateral side **162** then bows outward and then inward toward the heel end **154**, and terminates at a point where the lateral side **162** meets the longitudinal, center axis **152**.

It should be understood that numerous modifications may be apparent to those skilled in the art in view of the foregoing description, and individual components thereof, may be incorporated into numerous articles of footwear. Accordingly, aspects of the article of footwear **100** and components thereof, may be described with reference to general areas or portions of the article of footwear **100**, with an understanding the boundaries of the heel region **140**, the midfoot region **142**, the forefoot region **144**, the medial side **160**, or the lateral side **162** as described herein may vary between articles of footwear. However, aspects of the article of footwear **100** and individual components thereof, may also be described with reference to exact areas or portions of the article of footwear **100** and the scope of the appended claims herein may incorporate the limitations associated with these boundaries of the heel region **140**, the midfoot region **142**, the forefoot region **144**, the medial side **160**, or the lateral side **162** discussed herein.

Referring again to FIGS. 1 and 2, the sole structure **102** includes a chassis **106** that is attached to a bottom surface of each pod of the plurality of pods **124**. As best shown in FIG. 2, the chassis **106** is configured in an elongated fork structure. The elongated fork structure of the chassis **106** includes the chassis stem **108**, the first chassis leg **110**, and the second chassis leg **112**. The chassis stem **108** is positioned in the heel region **140** of the sole structure **102** and includes a heel joint **120** that connects the first chassis leg **110** and the second chassis leg **112**. The chassis stem **108** may be integral with the first chassis leg **110** and the second chassis leg **112**. The chassis stem **108** extends upwardly (i.e. away from the ground) from the heel joint **120**. In a particular embodiment, the chassis stem **108** extends upwardly, around at least a

portion of a pod **124** that is disposed in the heel region **140**. A portion of the chassis stem **108** may be connected to at least one pod **124**. In other embodiments, the entirety of the chassis stem **108** may be connected to one or more pods **124**.

The elongated fork structure of the chassis **106** includes the first chassis leg **110**. The first chassis leg **110** extends from the chassis stem **108**. The first chassis leg **110** extends along the bottom surface of the plurality pods **124**, and through the heel region **140**, the midfoot region **142**, and the forefoot region **144** of the sole structure **102**. The first chassis leg **110** defines at least a portion of the bottom-most surface of the article of footwear **100**, and may define at least a portion of the bottom-most surface of the article of footwear **100** in each of the heel region **140**, the midfoot region **142**, and the forefoot region **144**. In some embodiments, the first chassis leg **110** may be integral with the chassis stem **108**. Alternatively, the first chassis leg **110** may be formed separately from and coupled to the chassis stem **108**, in any suitable manner. The first chassis leg **110** may be disposed substantially on a medial portion of the sole structure **102**. Alternatively, the first chassis leg **110** may be disposed substantially on a lateral portion of the sole structure **102**.

The elongated fork structure of the chassis **106** also includes the second chassis leg **112**. The second chassis leg **112** may be substantially similar to the first chassis leg **110**. The second chassis leg **112** extends from the chassis stem **108**. The second chassis leg **112** is positioned opposite the first chassis leg **110**, such that if the first chassis leg **110** is disposed substantially on a medial portion of the bottom surface of the sole structure **102**, then the second chassis leg **112** is disposed substantially on a lateral portion of the sole structure **102**. In certain embodiments, the inverse arrangement of the first chassis leg **110** and the second chassis leg **112** is possible. Like the first chassis leg **110**, the second chassis leg **112** extends from the chassis stem **108** and through the heel region **140**, the midfoot region **142**, and the forefoot region **144** of the sole structure **102**. The second chassis leg **112** defines at least a portion of the bottom-most surface of the article of footwear **100**, and may define at least a portion of the bottom-most surface of the article of footwear **100** in each of the heel region **140**, the midfoot region **142**, and the forefoot region **144** of the sole structure. The second chassis leg **112** may be integral with the chassis stem **108**. In a particular embodiment, the entire chassis **106** (including at least the stem **108**, first leg **110**, and the second leg **112**) may be formed as a single unitary piece that is integrally formed from a single material. Alternatively, the second chassis leg **112** may be formed separately and coupled to the chassis stem **108**, in any suitable manner.

The chassis **106** may include the toe joint **122**. In embodiments that include the toe joint **122**, the toe joint **122** is positioned between the first chassis leg **110** and the second chassis leg **112** in the forefoot region **144** of the sole structure **102**. The toe joint **122** may be substantially similar to the portion of the chassis stem **108** that forms the heel joint **120** between the first chassis leg **110** and the second chassis leg **112** in the heel region **140** of the sole structure **102**. Alternatively, no toe joint **122** may be provided. In this instance, the first chassis leg **110** and the second chassis leg **112** remain distanced from each other throughout the entirety of the midfoot region **142** and the forefoot region **144**.

The elongated fork structure of the chassis **106** also includes the continuous elongated groove **114**. The continuous elongated groove **114** is positioned between the first chassis leg **110** and the second chassis leg **112**. The con-

tinuous elongated groove **114** has a first end **116** that is located in the heel region **140**, adjacent to the portion of the chassis stem **108** that forms the heel joint **120**. The continuous elongated groove **114** extends along substantially the entire length of the first chassis leg **110** and the second chassis leg **112**. The continuous elongated groove **114** separates substantially all of the length of the first chassis leg **110** from substantially all of the length of the second chassis leg **112**. The continuous elongated groove **114** may have second end **118** that is located in the forefoot region **144** of the sole structure **102**, adjacent to the toe joint **122**.

The second end **118** of the continuous elongated groove **114** may be adjacent to the toe joint **122**, when provided, such that the continuous elongated groove **114** is fully enclosed within and surrounded by the chassis **106** (in the longitudinal direction), as measured along the central, longitudinal axis **152**. Alternatively, the second end **118** of the continuous elongated groove **114** may be open/exposed in the forefoot region **144**, such that the chassis **106** does not fully enclose the length of the continuous elongated groove **114**. In one embodiment, the width of the continuous elongated groove **114** tapers proximate both the first end **116** and the second end **118**. In other embodiments, the width of the elongated groove **114** tapers at the first end **116** (in the heel region **140**) but remains substantially uniform at the second end **118** (in the forefoot region **144**). Alternatively, the width of the elongated groove **114** may remain substantially uniform at both the first end **116** and the second end **118**. Additionally, the continuous elongated groove **114** may be wider in the midfoot region **142** and/or forefoot region **144** than in the heel region **140**.

The continuous elongated groove **114** may be configured as an opening that can extend through the entire thickness of the chassis **106**. Alternatively, the continuous elongated groove **114** may be configured as an area that does not extend through the entire thickness of the chassis **106**, but instead forms a portion of the chassis **106** having a reduced thickness. The thickness of the chassis **106** may be reduced by any suitable amount, throughout the elongated groove **114**. For example, the thickness of the chassis **106** may be reduced by at least 50%, or at least 60%, or at least 70%, or at least 90%, or at least 95%, throughout the elongated groove **114**. In an embodiment, at least a portion of one or more of the pods may be visible through a portion of the continuous elongated groove **114**.

Still referring to FIGS. 1 and 2, the chassis **106** may also include additional optional components, such as a plurality of ground contacting regions **166** (which themselves may include traction pads), a heel support plate **168**, and/or a support fin **170**.

When present, the ground contacting regions **166** are disposed on portions of the bottom surface of the chassis **106**. Put another way, when an article of footwear **100** is worn by a user, portions of the bottom surface of the chassis **106** (i.e. ground contact regions **166**) directly interface with the ground on which the user walks or stands. The first chassis leg **110** and/or the second chassis leg **112** may comprise a plurality of ground contacting regions **166**. Specifically, both the first chassis leg **110** and the second chassis leg **112** may each comprise a plurality of ground contacting regions **166**. The chassis **106** may also comprise one or more traction pads. The traction pads may be positioned in some or all of the ground contacting regions **166**. The traction pads may be integrally formed with the chassis **106**. Alternatively, the traction pads may be coupled to the chassis **106** in any suitable manner. The ground contacting portions **166** of the traction pads, i.e. the bottom surface of

the traction pads, may include “high abrasion” textures or other features that are suitable for improving grip between the article of footwear **100** and the ground. The chassis **106** may be formed from one or more materials to impart durability, wear-resistance, abrasion resistance, or traction to the sole structure. For example, the chassis may be formed from rubber.

Still referring to FIGS. **1** and **2**, the chassis **106** may also include additional optional components, such as a plurality of ground contacting regions **166** (which themselves may include traction pads), a heel support plate **168**, and/or a support fin **170**.

When present, the heel support plate **168** that extends upwardly, i.e., away from the ground, from the chassis stem **108**. The heel support plate **168** may extend along and around at least a portion of at least one pod **124**. The heel support plate **168** may extend along at least some of the upper **104** in the heel region **140**. The heel support plate **168** may be attached to a portion of at least one pod **124**. Additionally, the heel support plate **168** may be distanced from, and therefore not attached directly to, the upper **104**. Beneficially, this may allow a heel portion of the upper **104** to have some space in which to expand, potentially to accommodate different foot sizes of different users, while simultaneously providing a region of rigidity and support to a heel portion of the upper **104**. The heel support plate **168** may further have a flange extending from the upper-most edge of the heel support plate **168**. The heel support plate **168** may have any suitable shape. The heel support plate **168** may have a substantially ovular or a substantially rectangular shape. The heel support plate **168** may be formed from any suitable material. For example, the heel support plate **168** may be formed from the same material as the chassis **106**. In a particular embodiment, the heel support plate **168** may be integral with the chassis **106**. Alternatively, the heel support plate **168** may be formed separately from and then subsequently attached to the chassis **106**.

When present, the support fin **170** extend upwardly from the first chassis leg **110** in the midfoot region **142** of the sole structure **102**. The support fin **170** may be positioned on lateral side **162** of the article of footwear **100**. Alternatively, the support fin **170** may be positioned on medial side **160** of the article of footwear **100**. Some articles of footwear may have multiple support fins **170**, such that a support fin **170** may be positioned on both the lateral side **162** and the medial side **160** of the article of footwear **100**. The support fin **170** may have any suitable shape. In a particular embodiment, the support fin **170** may have a substantially triangular shape. Alternatively, support fin **170** may have a shape that is commonly associated with the dorsal appendage of a dolphin or shark, i.e., a fin. The support fin **170** may be made from any suitable material. In an embodiment, the support fin **170** may be made from the same material as the chassis **106**. The support fin **170** may be integral with the chassis **106**. Alternatively, the support fin **170** may be formed separately from and subsequently attached to the chassis **106**. In certain embodiments, the support fin **170** may be attached to at least a portion of one or more pods **124**. Additionally, the support fin **170** may be distanced from, and therefore not attached directly to, the side of the upper **104**. Beneficially, this may allow the upper **104** to have space in which to expand, potentially to accommodate different foot sizes of different users, while simultaneously providing a region of rigidity and support to a side of the upper **104**.

Referring now to FIGS. **4** and **5**, a portion of the chassis **106** comprising the elongated fork structure is shown. The elongated fork structure provides the article of footwear **100**

numerous benefits. For example, the elongated fork structure provides an article of footwear **100** with improved flexibility. With particular reference to FIG. **4**, the elongated fork structure provides the article of footwear **100** with greater lateral/medial torsional flexibility, which may be particularly beneficial in athletic shoes. Referring now to FIG. **5**, the elongated fork structure provides the article of footwear **100** with greater flexibility for movement in the vertical direction in the forefoot region **144**, while simultaneously providing greater rigidity and support in the heel region **140**. Additionally, the elongated fork structure of the chassis **106** according to the present disclosure may have relatively few components, may require relatively few materials to manufacture, and reduce assembly time; all of which may beneficially reduce the cost of manufacturing related footwear.

Turning back to FIGS. **1** and **2**, the sole structure **102** includes the plurality of pods **124**. The plurality of pods **124** have at least a first pod **126**, a second pod **128**, and third pod **130**. The chassis **106** is coupled with the plurality of pods **124**. The plurality of pods **124** is coupled with the upper **104**.

Each pod in the plurality of pods **124** may have two portions: (1) a pod body, and (2) an optional coating. As used herein, the term “pod body” refers to all portions of the pod **124** other than an optional coating. The pod body forms at least the significant majority of the mass of the pod **124** and provides the pod **124** with its general shape. In pods **124** that have an optional coating, the coating is disposed around the outside surface of its pod body. In pods **124** that do not have a coating, the pod body makes up the entirety of the pod **124**, i.e., the pod **124** consists only of a pod body. For these pods **124**, the terms “pod body” and “pod” become synonymous and can be used interchangeably.

Each pod **124** is formed from a foam material. More specifically, each pod body is formed from a foam material. The foam material(s) may include any suitable compositions. For example, foam material may be constructed from ethylene-vinyl acetate (EVA), interpolymers that include EVA, or a similar type of material. In some embodiments, the foam material may be an EVA-Solid-Sponge (“ESS”) material, an EVA foam (e.g., PUMA® ProFoam Lite™, IGNITE Foam), polyurethane, polyether, an olefin block copolymer, a thermoplastic material (e.g., a thermoplastic polyurethane, a thermoplastic elastomer, a thermoplastic polyolefin, etc.), or a supercritical foam. The foam material may comprise only a single polymeric material or may comprise a blend of materials, such as an EVA copolymer, a thermoplastic polyurethane, a polyether block amide (PEBA) copolymer, and/or an olefin block copolymer.

A foam material may be formed by a supercritical foaming process, such that the foam material form a supercritical foam. The supercritical foam may have micropore foams or particle foams, such as a TPU, EVA, PEBAX®, or mixtures thereof, manufactured using a process that is performed within an autoclave, an injection molding apparatus, or any sufficiently heated/pressurized container that can process the mixing of a supercritical fluid (e.g., CO₂, N₂, or mixtures thereof) with a material (e.g., TPU, EVA, polyolefin elastomer, or mixtures thereof) that is preferably molten. During an exemplary process, a solution of supercritical fluid and molten material is pumped into a pressurized container, after which the pressure within the container is released, such that the molecules of the supercritical fluid rapidly convert to gas to form small pockets within the material and cause the material to expand into a foam, which may be used in a pod of the present disclosure. In some embodiments, pods of the present disclosure may be formed using alternative methods known in the art, including the use of an expansion press, an

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injection machine, a pellet expansion process, a cold foaming process, a compression molding technique, die cutting, or any combination thereof. For example, a pod **124** may be formed using a process that involves an initial foaming step in which supercritical gas is used to foam a material and then compression molded or die cut to a particular shape.

Some pod bodies may consist only of a single foam material. However, in some pods **124** each individual pod body may comprise a combination of multiple foam materials.

In pods **124** having an optional coating, the optional coating may be a skin layer disposed around the outside of the pod body. Alternatively, the coating may be added to the outside of the pod body by spraying, extrusion coating, painting, dipping, or any other suitable method. The coating may be made of any materials suitable for a particular application. For example, the coating may include a foam material as described here, where the foam material of the coating is different than the foam material of the pod body. The coating may alter or improve a target property of the pod **124**. For example, the coating may provide the pod **124** with any or all of: a harder outer surface, an improved resistance to water, an improved resistance to stains, a higher melting temperature, a higher flashpoint, better bonding properties, and/or a preferable appearance.

A sole structure **102** may have pods **124** made from a number of different foam materials. For example, one pod **124** may be made of a foam material that is different from the foam material of any of the other pods **124**. Further, a sole structure may have a plurality of pods **124** in which every pod **124** is made from a foam material that is different than the foam material of every other pod **124**. Put another way, in such an embodiment, each pod **124** is made of a unique foam material. For example, in the article of footwear **100** shown in FIGS. **1** and **2**, the first pod **126** may be formed from a first foam material, the second pod **128** may be formed from a different second foam material, and the third pod **130** may be formed from still another different third foam material.

It is often especially useful if the different foam materials used to make a plurality of pods **124** vary in density, hardness, and/or elasticity. Referring now to FIG. **6**, a weight **250** and three different types of foam material, formed into wedge shapes (**252**, **254**, and **256** respectively), are shown. The weight **250** is shown resting on the top surface of each wedge (**252**, **254**, and **256** respectively) of foam material. The foam material in wedge **252** is less dense than the foam material in wedge **254**, which in turn is less dense than the foam material in wedge **256**. The differences in density cause the weight **250** to compress the wedge **252** to a greater degree than it does the wedge **254**. The weight **250** compresses the wedge **256** even less than it does the wedge **254**, the weight appearing to have little if any compressive impact upon wedge **256**. As such, the foam material of wedge **256** may be particularly useful in components of an article of footwear that tend to be load bearing and therefore need to provide support, whereas the foam material of wedge **252** may be particularly useful in components of an article of footwear that tend to experience high amounts of impact force and therefore need to provide cushioning. A particular foam material may have a density within the range between about 0.05 grams per cubic centimeter (g/cm³) and about 0.50 g/cm³, or between about 0.20 g/cm³ and about 0.40 g/cm³. The foam material of particular pods may have a Shore A hardness between about 45 and about 60.

Further, the sole structure **102** may have a plurality of pods **124** in which some of the pods **124** form a group,

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wherein each pod **124** in the group is made of the same foam material. Some sole structures **102** may have pods **124** that form multiple groups, wherein the pods **124** in a given group are all made of a single foam material, but every group uses a different foam material. For example, a plurality of pods **124** may be formed from three foam materials (**1**, **2**, and **3**), such that the pods **124** form three groups (A, B, and C), wherein the pods **124** of Group A all use foam material **1**, the pods **124** of Group B all use foam material **2**, and the pods **124** of Group C all use foam material **3**.

The pods **124** in a group may all be positioned near each other, may be located in multiple clusters, or may be spread randomly throughout a sole structure. It may be beneficial to cluster certain groups of pods **124** together, in particular areas of the sole structure. For example, it may be useful to position a group of pods **124** made from a foam material having certain properties in the forefoot region **144**, while clustering other pods **124** that are made from a foam material having different properties in the heel region **140**, so that different parts of a single sole structure can provide different amounts of properties such as support, cushioning, impact absorption, and/or stain resistances, as may be useful. It may be beneficial to position pods **124** that provide greater amounts of support in the heel region **140**, whereas it may be beneficial to position pods **124** with greater impact absorption in the forefoot region **144**. A sole structure **102** may comprise any number of such groups of pods **124** (each group made of a single foam material or combination of foam materials), positioned throughout the sole structure **102** using any suitable design.

Positioning of pods **124** having certain properties that are particularly suitable to certain regions of a sole structure may also occur with single pods **124**. When only a single pod **124** is positioned in a particular region of a sole structure, it is typically larger than pods **124** that are positioned in groups. For example, in one embodiment, a single pod **124** may occupy, e.g., contact directly or indirectly, at least 50% of the surface area of the bottom surface of the upper **104** in a given region. In another embodiment, a single pod **124** may occupy at least 70%, or at least 80%, or at least 90% of the surface area of the bottom surface of the upper **104** in a given region. Additionally, a single pod **124** may occupy substantially all of the surface area of the bottom surface of the upper **104** in a given region.

FIGS. **1** and **2** depict an embodiment using larger pods **124**, as described above. Each pod in the plurality of pods **124** (i.e. the first pod **126**, the second pod **128**, and the third pod **130**) has a substantially elongated shape. Additionally, each pod in the plurality of pods **124** is positioned in a particular (and different) region of the article of footwear **100**. The first pod **126** is positioned in the heel region **140**, the second pod **128** is positioned in the midfoot region **142**, and third pod **130** is positioned in the forefoot region **144**. Each pod **124** runs along substantially the entire longitudinal length of its respective region, as measured along longitudinal, central axis **152**. With particular reference to FIG. **2**, each pod **124** also runs across substantially the entire width of the article of footwear **100**, from the lateral side **162** to the medial side **160**, in its respective region.

The first, second, and third pods (**126**, **128**, **130**) have different densities such that: 1st pod density > 2nd pod density > 3rd pod density. The first pod **126** is positioned substantially in a heel region **140**, the second pod **128** is positioned substantially in a midfoot region **142**, and the third pod **130** is positioned substantially in a forefoot region **144**. This arrangement is a particularly beneficial way to provide superior support to a user's foot in the heel region,

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which often receives a disproportionately large amount of a user's weight, while also providing superior impact absorption in the forefoot region, which often experiences a disproportionately large amount of impact force during use, such as running.

Referring now to FIG. 7, a plurality of pods **124** are shown, the pods **124** disposed on the bottom surface of a skeletal representation of a foot of a user. The plurality of pods **124** includes four pods, a first pod **210**, a second pod **212**, a third pod **214**, and a fourth pod **216**. The first pod **210** has a substantially cylindrical shape and is disposed proximate the heel of the user's foot, i.e. in a heel region. The second pod **212** has a substantially elongated shape and is disposed along a medial portion of a midfoot region of the user's foot. The third pod **214** also has a substantially elongated shape and is disposed along lateral portions of the midfoot and forefoot regions of a user's foot. The fourth pod **216** has a substantially elongated shape and is disposed along a medial portion of the forefoot region of a user's foot. A plurality of pods may have 5 pods, 6 pods, 7 pods, 8 pods, 9 pods, 10 pods, or any number of pods suitable for a particular application. The first pod **210** may be formed from a relatively dense foam material (such as the foam material of wedge **256**) to provide superior support, whereas the fourth pod **216** may be formed from a foam material with relatively low density (such as the foam material of wedge **252**) to provide superior cushioning. The second pod **212** and the third pod **214** may be formed from one or more foam material(s) have a mid-range density (in comparison to the foam material of the first pod **210** and the fourth pod **216**) to provide a balance of support/cushioning properties. The second pod **212** and the third pod **214** may be formed from the same foam material. Alternatively, the second pod **212** and the third pod **214** may be made from different foam materials.

Referring back to FIGS. 1 and 2, the upper **104** is shown disposed above and coupled with the sole structure **102**. The upper **104** may be attached to the sole structure via adhesive or stitching. The upper **104** could be formed conventionally from multiple elements, e.g., textiles, polymer foam, polymer sheets, leather, or synthetic leather, which are joined through bonding or stitching at a seam. In some embodiments, the upper **104** of the article of footwear **100** is formed from a knitted structure or knitted components. In various embodiments, a knitted component may incorporate various types of yarn that may provide different properties to an upper. For example, an upper mesh layer may be warp knit, while a mesh backing layer may comprise a circular knit.

The upper **104** extends along the lateral side **162** and the medial side, and across the heel region **140**, the midfoot region **142**, and the forefoot region **144** to house and enclose a foot of a user. When fully assembled, the upper **104** also includes the interior surface **172** and the exterior surface **174**. The interior surface **172** faces inward and generally defines the interior cavity **176**, and the exterior surface **174** of the upper **104** faces outward and generally defines an outer perimeter or boundary of the upper **104**. The interior surface **172** and the exterior surface **174** may include portions of the upper layers disclosed above. The upper **104** also includes an opening that is at least partially located in the heel region **140** of the article of footwear **100**, that provides access to the interior cavity **176** and through which a foot may be inserted and removed.

In some embodiments, various layers of the upper **104** are heat pressed together so as to bond the various layers of the upper **104**. For example, layers that comprise the upper **104** can be heat pressed together all at once and at a single

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temperature. Various layers of the upper **104** may be waterproof or semi-waterproof, and may include a plurality of layers of mesh or other materials. The materials that comprise the upper **104** may include an inner mesh layer, a thermoplastic polyurethane (TPU) film, and an outer mesh layer. A TPU skin may be applied along the outer surface of the upper.

With reference to the material, or materials, that form the upper **104**, the specific properties that a particular type of yarn will impart to an area of a knitted component may at least partially depend upon the materials that form the various filaments and fibers of the yarn. For example, cotton may provide a soft effect, biodegradability, or a natural aesthetic to a knitted material. Elastane and stretch polyester may each provide a knitted component with a desired elasticity and recovery. Rayon may provide a high luster and moisture absorbent material, wool may provide a material with an increased moisture absorbance, nylon may be a durable material that is abrasion-resistant, and polyester may provide a hydrophobic, durable material.

Other aspects of a knitted component may also be varied to affect the properties of the knitted component and provide desired attributes. For example, a yarn forming a knitted component may include monofilament yarn or multifilament yarn, or the yarn may include filaments that are each formed of two or more different materials. In addition, a knitted component may be formed using a particular knitting process to impart an area of a knitted component with particular properties. Accordingly, both the materials forming the yarn and other aspects of the yarn may be selected to impart a variety of properties to particular areas of the upper **104**.

An elasticity of a knit structure may be measured based on comparing a width or length of the knit structure in a first, non-stretched state to a width or length of the knit structure in a second, stretched state after the knit structure has a force applied to the knit structure in a lateral direction.

In some instances, other elements, e.g., plastic material, logos, trademarks, etc., may also be applied and fixed to an exterior surface using glue or a thermoforming process. The properties associated with an upper, e.g., a stitch type, a yarn type, or characteristics associated with different stitch types or yarn types, such as elasticity, aesthetic appearance, thickness, air permeability, or scuff-resistance, may be varied.

The article of footwear **100** may also include a tightening system, such as a system that includes a lace and a plurality of eyelets. The tightening system may be a hook-and-loop fastening system, such as Velcro®. Alternatively, the tightening system may be any other laceless fastening system known in the art. Still further, the tightening system may include a different manual lacing system, a rotary closure device, or an automatic lacing system, such as the lacing systems described in U.S. patent application Ser. No. 15/780,368, filed on May 31, 2018, now issued as U.S. Pat. No. 10,758,011, and U.S. patent application Ser. No. 16/392,470, filed on Apr. 23, 2019, now issued as U.S. Pat. No. 11,033,079, both of which are hereby incorporated by reference in their entirety.

Referring now to FIGS. 8 and 9, a particular article of footwear **300** is shown. The article of footwear **300** is substantially similar to the article of footwear **100** (with corresponding elements having corresponding reference numerals, in the **300** series). For example, the article of footwear **100** has a sole structure **102** and an upper **104** and the article of footwear **300** has a sole structure **302** and an upper **304**. Like the sole structure **102**, the sole structure **302** also includes a chassis **306** having a chassis stem **308**, a first chassis leg **310**, a second chassis leg **312**, and a continuous

elongated groove 314. The continuous elongated groove 314 has a first end 316, a second end 318, a heel joint 320, and a toe joint 322. The sole structure 302 also includes a plurality of pods 324. The chassis 306 is coupled with the plurality of pods 324. The plurality of pods 324 is connected directly with the upper 304. The primary difference between article of footwear 100 and article of footwear 300 is that the sole structure 302 encapsulates a plurality of pods 324. As used herein the term “encapsulate” means to surround most or all of an object. For example, a case that surrounds an object on substantially all sides but which has an opening/aperture, would still be said to encapsulate the object. The sole structure 302 also includes a plurality of capsule(s) 372 that surround(s) the pods 324. The article of footwear 300 has a heel region 340, a midfoot region 342, and a forefoot region 344 and that are substantially similar to the heel/midfoot/forefoot regions of the article of footwear 100 described above. The article of footwear 300 also has a medial side (not shown) and a lateral side 362 that are substantially similar to the medial/lateral sides of the article of footwear 100 described above.

The continuous elongated groove 314 is disposed such that it does not extend through the entire thickness of the chassis 306 but rather comprises a reduced thickness region (as described in more detail above) of the chassis 306 along substantially the entire length of the continuous elongated groove 314.

The chassis 306 encapsulates each pod in the plurality of pods 324. The chassis 306 may encapsulate each pod 324 within its own capsule 372. Alternatively, the chassis 306 could encapsulate all of the pods 324 in a single capsule 372. In some implementations, the chassis 306 may encapsulate some pods 324 that are located proximate each other into a joint capsule, such that there are multiple joint capsules that each contain multiple pods within a single sole structure 302. In particular embodiments, the chassis 306 encapsulates each of the pods in the plurality of pods 324 without retaining any fluid, including any air, within any of the capsules 372.

The chassis 306 may also include additional optional components, such as a plurality of flex grooves. The flex grooves 374 may be disposed on lateral side 362 of the article of footwear 300, on the medial side (not shown) of the article of footwear 300, or both. The plurality of flex grooves 374 may include any number of individual flex grooves 374 suitable. In the embodiment shown in FIG. 8, the chassis 306 has a plurality of flex grooves 374 that includes four individual flex grooves 374. Beneficially, the plurality of flex grooves 374 provide the article of footwear 300 with added flexibility, which may be particularly important in embodiments that employ a chassis 306 having a greater thickness. For example, a plurality of flex grooves 374 may be particularly useful in embodiments having a sole structure 302 that includes a chassis 306 that encapsulates a plurality of pods 324, as such encapsulation may tend to increase the thickness of the sole structure 302.

Still referring to FIGS. 8 and 9, at least a portion of the chassis 306 comprises a translucent material. In some embodiments, at least the portion of the chassis 306 that encapsulates the plurality of pods 324, i.e., the portion(s) of the chassis 306 that form the capsules 372, is made from a translucent material, such that the pods are visible through the chassis 306. As shown in FIGS. 8 and 9, portions of the chassis 306 are made from a translucent material, such that the pods can be seen through a lateral side 362 of the chassis 306 (as shown in FIG. 8) and through portions of the bottom surface of the chassis 306 (as shown in FIG. 9).

Moving now to FIGS. 10 and 11, an article of footwear 400 is shown. The article of footwear 400 is substantially similar to the article of footwear 300 (with corresponding elements having corresponding reference numerals, in the 400 series). For example, the article of footwear 300 has a sole structure 302 and an upper 304, and the article of footwear 400 has a sole structure 402 and an upper 404. Like the sole structure 302, the sole structure 402 also includes a chassis 406 having a chassis stem, a first chassis leg, a second chassis leg, and a continuous elongated groove 414. The sole structure 402 also includes a plurality of pods 424. The chassis 406 is coupled with the plurality of pods 424. The plurality of pods 424 is connected directly with the upper 404. The primary difference between article of footwear 300 and article of footwear 400 is that the article of footwear 400 includes a sole structure 402 with a chassis 406 that is formed from a “soft chassis material” and also has hard chassis coverings 476. Similar to the chassis 306 of the article of footwear 300, described above, the chassis 406 encapsulates the plurality of pods 424. The “soft chassis material” and/or the hard chassis coverings 476 may be formed from a translucent material. The article upper 404 may be similar to the upper 104, described above. The article of footwear 400 has a heel region 440, a midfoot region 442, and, a forefoot region 444 that are substantially similar to the heel/midfoot/forefoot regions described above, with reference to the article of footwear 100.

The soft chassis material may be any suitable material, for example, the soft chassis material may be a rubber (such as silicon rubber) or a thermoplastic elastomer (TPE) such as styrenic block copolymers, polyamides, or polyurethanes, or other similar materials. The hard chassis covering 476 may be any suitable material, for example, the hard chassis covering 476 may be polyvinyl chloride (PVC), polyethylene terephthalate (PET), or high-density polyethylene, or other similar materials.

The hard chassis covering 476 may include ground contacting regions 466. Ground contacting regions 466 may also comprise traction pads, as described above. In some embodiments, the hard chassis covering 476 may comprise portions of a ground contacting region 466, such that hard chassis covering 476 partially or completely surrounds one or more traction pads. In some embodiments, portions of a ground contacting region 466 comprise a traction pad that is not surrounded by the hard chassis covering 476. For example, as best shown in FIG. 11, a portion of the hard chassis covering 476 comprises a portion of a ground contacting region 466 that completely surrounds traction pad 466a. However, another ground contacting region 466 comprises traction pad 466b, which is not surrounded by the hard chassis covering 476.

Any of the embodiments described herein may be modified to include any of the structures or methodologies disclosed in connection with different embodiments. Similarly, materials or construction techniques other than those disclosed above may be substituted or added in some embodiments according to known approaches. Further, the present disclosure is not limited to articles of footwear of the type specifically shown. Still further, aspects of the articles of footwear of any of the embodiments disclosed herein may be modified to work with any type of footwear, apparel, or other athletic equipment.

As noted previously, it will be appreciated by those skilled in the art that while the disclosure has been described above in connection with particular embodiments and examples, the disclosure is not necessarily so limited, and that numerous other embodiments, examples, uses, modifications and

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departures from the embodiments, examples and uses are intended to be encompassed by the claims attached hereto.

We claim:

1. An article of footwear, comprising:
an upper; and
a sole structure coupled with the upper, the sole structure defining a forefoot region, a midfoot region that corresponds with a portion of the article of footwear that is configured to encase an arch of a foot, and a heel region that is proximate and adjoining the midfoot region, the sole structure comprising:
a plurality of pods, each pod having a pod body and optionally having a coating disposed on an outer surface of the pod body; and
a chassis attached to a bottom surface of each pod in the plurality of pods,
wherein the pod body of each pod in the plurality of pods comprises a foam material,
wherein the chassis comprises a plurality of ground contacting regions, and
wherein the chassis has an elongated fork structure, the elongated fork structure comprising:
a chassis stem;
a first chassis leg extending from the chassis stem through the heel region, the midfoot region, and the forefoot region of the sole structure;
a second chassis leg extending from the chassis stem through the heel region, the midfoot region, and the forefoot region of the sole structure; and
a continuous elongated groove positioned between the first chassis leg and the second chassis leg, the continuous elongated groove extending along substantially the entire length of the first chassis leg and the second chassis leg,
wherein the upper has a bottom surface that defines a bottom surface area and each pod in the plurality of pods has a top surface, wherein a portion of the top surface of each pod in the plurality of pods is attached to a portion of the bottom surface of the upper, and wherein in a majority of the bottom surface area, the bottom surface of the upper contacts the top surface of a pod in the plurality of pods.
2. The article of footwear of claim 1, wherein the pod body of each pod in the plurality of pods consists of the foam material.
3. The article of footwear of claim 2, wherein the foam material in the pod body of at least one pod in the plurality of pods has a density that is different from a density of the foam material in the pod body of at least one other pod in the plurality of pods.
4. The article of footwear of claim 3, wherein the foam material in the pod body of each pod in the plurality of pods has a density that is different from the density of the foam material in the pod body of every other pod in the plurality of pods.
5. The article of footwear of claim 1, wherein the plurality of pods comprises at least a first pod, a second pod, and a third pod, and
wherein a density of the foam material in the pod body of the first pod is greater than a density of the foam material in the pod body of the second pod,
wherein the density of the foam material in the pod body of the second pod is greater than a density of the foam material in the pod body of the third pod, and
wherein the first pod is positioned substantially in the heel region of the sole structure, the second pod is positioned substantially in the midfoot region of the sole

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- structure, and the third pod is positioned substantially in the forefoot region of the sole structure.
6. The article of footwear of claim 1, wherein the chassis has an elongated fork structure, the chassis is not directly attached to the upper.
 7. The article of footwear of claim 1, wherein the chassis comprises a support fin extending upward from the first chassis leg in the midfoot region of the sole structure.
 8. An article of footwear, comprising:
an upper; and
a sole structure coupled with the upper, the sole structure defining a forefoot region, a midfoot region, and a heel region, the sole structure comprising:
a plurality of pods, each pod having a pod body and optionally having a coating disposed on an outer surface of the pod body; and
a chassis attached to a bottom surface of each pod in the plurality of pods,
wherein the pod body of each pod in the plurality of pods comprises a foam material,
wherein the foam material in the pod body of at least one pod in the plurality of pods has a density that is different than a density of the foam material in the pod body of at least one other pod in the plurality of pods, wherein the chassis comprises a plurality of ground contacting regions,
wherein the chassis is a single unitary piece that is integrally formed from a single material, and
wherein the chassis comprises a heel support plate extending upward from a chassis stem.
 9. The article of footwear of claim 8, wherein the foam material in the pod body of each pod in the plurality of pods has a density that is different from the density of the foam material in the pod body of every other pod in the plurality of pods.
 10. The article of footwear of claim 8, wherein the plurality of pods comprises at least a first pod, a second pod, and a third pod, and
wherein a density of the foam material in the pod body of the first pod is greater than a density of the foam material in the pod body of the second pod,
wherein the density of the foam material in the pod body of the second pod is greater than a density of the foam material in the pod body of the third pod, and
wherein the first pod is positioned substantially in the heel region of the sole structure, the second pod is positioned substantially in the midfoot region of the sole structure, and the third pod is positioned substantially in the forefoot region of the sole structure.
 11. The article of footwear of claim 8, wherein the upper has a bottom surface and each pod in the plurality of pods has a top surface, and
wherein the top surface of each pod in the plurality of pods is attached to the bottom surface of the upper.
 12. The article of footwear of claim 11, wherein the bottom surface of the upper defines a bottom surface area, and
wherein in a majority of the bottom surface area, the bottom surface of the upper contacts the top surface of a pod in the plurality of pods.
 13. The article of footwear of claim 11, wherein the elongated fork structure comprises:
a first chassis leg extending from the chassis stem through the heel region, the midfoot region, and the forefoot region of the sole structure;

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a second chassis leg extending from the chassis stem through the heel region, the midfoot region, and the forefoot region of the sole structure; and
 a continuous elongated groove positioned between the first chassis leg and the second chassis leg, the continuous elongated groove extending along substantially the entire length of the first chassis leg and the second chassis leg.

14. An article of footwear, comprising:

an upper; and

a sole structure coupled with the upper, the sole structure defining a forefoot region, a midfoot region, and a heel region, the sole structure comprising:

a plurality of pods, each pod having a pod body and optionally having a coating disposed on an outer surface of the pod body; and

a chassis attached to a bottom surface of each pod in the plurality of pods,

wherein the pod body of each pod in the plurality of pods comprises a foam material,

wherein the foam material in the pod body of at least one pod in the plurality of pods has a density that is different than a density of the foam material in the pod body of at least one other pod in the plurality of pods,

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wherein the chassis at least partially encapsulates each pod in the plurality of pods,

wherein the chassis comprises a plurality of ground contacting regions, and

wherein the chassis is not directly attached to the upper.

15. The article of footwear of claim **14**, wherein the chassis is indirectly coupled to the upper by way of the plurality of pods.

16. The article of footwear of claim **14**, wherein the chassis has an elongated fork structure, the elongated fork structure comprising:

a chassis stem;

a first chassis leg extending from the chassis stem through the heel region, the midfoot region, and the forefoot region of the sole structure;

a second chassis leg extending from the chassis stem through the heel region, the midfoot region, and the forefoot region of the sole structure; and

a continuous elongated groove positioned between the first chassis leg and the second chassis leg, the continuous elongated groove extending along substantially the entire length of the first chassis leg and the second chassis leg.

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